

Case (0,0) M_A, M_B in cutoff, M_{L_1} in saturation.

$$\boxed{V_{od} = 5.09V}$$

 M_C is in triode, M_{L_2} in saturation

$$I_{M_{L_2}} = 0.5 (2(V_T - V_T) - (-0.3))^2 = 0.045 \text{ mA}$$

$$I_{M_C} = 0.5 (2(5.09 - 0.55)V_T - V_T^2) = 0.045$$

$$\therefore V_T = 9.07, 0.01V$$

$$\boxed{\therefore V_T = 0.01V}$$

$$I = 0.045$$

$$P = (5.09 - 0) \times 0.045 = 0.22905 \text{ mW}$$

Case (0,1) M_A in cutoff, M_B in triode, M_{L_1} satn.

$$I_{M_B} = 0.5 (2(5.09 - 0.6)V_{od} - V_{od}^2)$$

$$I_{M_C} = 0.5 (5.09 - V_{od} - 0.1)^2$$

$$\text{We get } V_{od} = 4.46V, 1.86V$$

$$\therefore V_{od} = 1.86V$$

M_1 is in triode, M_{L2} in saturation.

so current will be same.

Only difference is, here we will have driver current as well.

$$P = (5.09 - 0) \times 4.89845 + 0.22905$$

$$= 25.1622 \text{ mW}$$

Case (1,0)

Solve similarly to get

$$\boxed{V_{od} = 1.82 \text{ V}}, I = 5.02445.$$

$$\therefore P = 25.804 \text{ mW}$$

Case (1,1)

$$I_{M_{L1}} = I_{M_A} + I_{M_B}$$

$$\Rightarrow 0.5(5.09 - 0.1 - V_{od})^2 = \overset{0.5 \times}{(2(5.09 - 0.6)V_{od} - V_{od}^2)} + 0.5(2(5.09 - 0.5)V_{od} - V_{od}^2)$$

$$\therefore V_{od} = 0.573 \text{ V}$$

$$P = 49.882 \text{ mW}$$

$$\text{Avg Power} = 25.269 \text{ mW}$$